

Utility of Field Weakening and Field-Oriented Control in Permanent-Magnet Synchronous Motors: A Case Study [†]

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Abstract: The significance of electric vehicles is progressively escalating, underscoring the criticality of technologies underpinning the functionality of their propulsion systems. This particular case study delves into the simulation of field-oriented control, coupled with field weakening, aimed at regulating a salient-pole permanent-magnet synchronous motor (PMSM). This approach involves the utilization of two cascaded loops for current and voltage, each employing PI controllers. The fine-tuning of these controllers' parameters hinges on the motor characteristics, as well as the desired response bandwidth.

Keywords: PMSM; field weakening; BLAC; FOC; SVPWM



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1. Introduction

Permanent-magnet synchronous motors (PMSMs) have gained prominence in electric mobility due to their impressive power density and capacity to surpass nominal speeds, necessitating the implementation of flux-weakening techniques, [1,2]. Research highlighted in [3] demonstrates that integrating a PMSM into an electric vehicle elevates its autonomy by 15% compared to an induction motor with equivalent power ratings. Articles such as [4–7] indicate that with the right control strategy, PMSM can provide a high torque-to-current ratio, a high power-to-weight ratio, and high efficiency and robustness.

This work introduces a comprehensive case study focused on a nested control architecture based on a field-oriented controller (FOC) tailored for a salient-pole permanent-magnet synchronous motor with a brushless alternating current (PMSM-BLAC), and a speed control loop. This case study provides details of a simulation that it is open to the community, using real specifications from a commercial motor. To evaluate the FOC controller design, a detailed model of the electrical and mechanical motor components expressed in a dq0 reference frame is proposed; in addition, the controller is tuned using a dedicated approach that is used in real synchronous-based motor systems of BLAC type so that we can produce as realistic a simulation as possible for further real implementation. The controller along the power electronics and the motor are mounted on a MATLAB/SIMULINK, where a space-vector pulse width modulation signal (SVPWM) drives the inverter, and field weakening is emulated to achieve speeds higher than the nominal one. The controller plus the motor system are evaluated in different scenarios, like locked rotor tests, speed and current control with different variable velocity profiles, experiments with different load–velocity profiles, and system evaluations with field weakening in constant load and torque conditions. The following sections present details about the controller and motor modelling methodology, experiments, results, and conclusions.

where p is the pole pair number, and R is the resistance of the stator windings. Recall that the inductances in the $dq0$ frame, L_q and L_d , are the same and equal to L for the PMSM-BLAC motor [8–15]. The remaining terms belongs to the magnetic torque T_e , mechanical and electrical velocity ω_m , ω_e , respectively, load torque T_m , coefficient of viscous friction B , and motor inertia J .

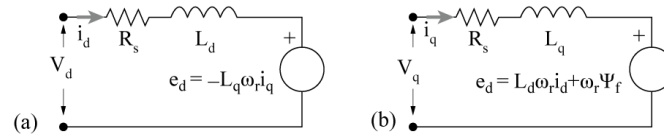


Figure 2. Equivalents of a PMSM motor on: (a) d axis and (b) q axis.

In order to analyse the three-phase stator voltage behaviour in the dq frame, we must depart from $\alpha\beta$ coordinates using a transform matrix that holds peak convention and a-phase-to-d-axis alignment [9]. The notation $^B_A(\cdot)$ stands for a transform matrix that maps a value from the A to B reference; in this case, it maps from abc to dq , yielding

$$^{dq}_{abc}C = ^{dq}_{\alpha\beta}C_{\alpha\beta}^{abc}C = \frac{2}{3} \begin{bmatrix} \cos(\theta) & \cos(\theta - 120^\circ) & \cos(\theta + 120^\circ) \\ -\sin(\theta) & -\sin(\theta - 120^\circ) & -\sin(\theta + 120^\circ) \end{bmatrix} \quad (5)$$

Recall that the implemented MATLAB model makes use of Equations (1)–(4), and the currents were transformed from a static reference frame abc to rotational axes $dq0$ using the transform matrix stated in Equation (5).

2.2. Vectorial Field-Oriented Control

In this work, we implemented a vectorial field-oriented control (FOC) scheme due to its excellent dynamics to address, with great accuracy, transient and steady-state responses [8,16,17]. The main objective of vectorial control relies on controlling the space vectors concerning the stator current [18,19]; furthermore, it allows for the de-coupling of speed and torque to achieve independent control of both variables.

Recall that in an AC motor, the space angle between the rotating stator field and the rotor flux changes due to the load, which causes an oscillatory response. In contrast, in a DC motor, the armature current directly controls the torque and the rotor field current, and they can be accessed independently. The angle between these variables is held orthogonally through a mechanical switching system, such as brushes and commutators.

The vectorial FOC emulates the performance of a DC machine because it observes the position of the rotor field and directs the stator field to achieve a constant 90-degree angle between both of them. The last condition is used to reach the maximum torque while we control the rotor speed independently. FOC control requires a position sensor to know the rotor angular pose at all time, which is also related to the rotor flux. It is worth mentioning that the stator field is directed through the three-phase current's phases [20–23]. In order to figure the open-loop current model out, and represent it on the $dq0$ axes, we depart from Equation (1), which allows us to determine the motor inductances as follows:

$$L_d \frac{di_d}{dt} = v_d - R_s i_d + \omega_e L_q i_q \quad \& \quad L_q \frac{di_q}{dt} = v_q - R_s i_q - \omega_e (L_d i_d + \Psi_m), \quad (6)$$

from which we can obtain the stator currents in $dq0$ and translate them into the Laplace domain as follows:

$$i_d(s) = \frac{V_d + \omega_e L_q i_q}{L_d s + R_s} \quad \& \quad i_q(s) = \frac{V_q - \omega_e L_d i_d - \omega_e \Psi_m}{L_q s + R_s} = \frac{V_q - \omega_e L_d i_d - \omega_e k_e}{L_q s + R_s}. \quad (7)$$

The torque can be derived from Equation (2), and because $L_d = L_q$, the PMSM-BLAC motor yields $T_e = \frac{3p}{2} \psi_m i_q = k_t \times i_q$. The motor's angular speed is computed from Equation (4), which can be expressed in the frequency domain as

$$\omega_m(s) = \frac{T_e - T_m}{Js + B}. \quad (8)$$

The above-mentioned equations are placed together in the simulation diagram showed in Figure 3, where we can notice the correlation between currents on the d and q axes, which forces us to implement a decoupling technique for implementing independent current controllers.

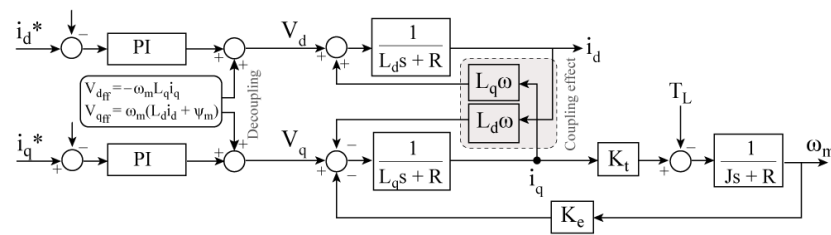


Figure 3. Simulation diagram of the PMSM-BLAC motor on $dq0$ axis.

The currents i_d and i_q are decoupled using a feedforward approach with EMF compensation [9,20–23], as described in the blocks of Figure 3, which will result in independent controlled current variations, where i_d is set to 0, and i_q varies according to the torque [24]. The current control selected in this work was a PI, due to its fast response to changes in the setpoint, its capability of reducing the error in steady state to zero [25–27], and the fact that it is the most common controller used at an industrial level. The PI control is modelled with no relation to any current i and its desired value noted as i^* , which can be expressed as a first-order system in the Laplace domain to reduce overshooting and oscillations, leading to

$$\frac{i(s)}{i^*(s)} = \frac{K_p s + K_i}{Ls^2 + (K_p + R)s + K_i} \approx \frac{W_{cc}}{s + W_{cc}}, \quad (9)$$

where W_{cc} denotes the cutoff frequency in the closed loop of the current controller. The tuning approach selected to compute the constants k_p and k_i was the principle of the optimal magnitude criterion, introduced by Sartorius and Oldenbourg. This principle states that a controller should have a closed-loop magnitude in the frequency domain as close to the unit as possible, in the widest possible frequency range [25–27]. In this sense, the controller constant can be computed as follows:

$$K_{i,c} = RW_{cc}, \text{ \& } K_{p,c} = LW_{cc}. \quad (10)$$

2.3. Speed Control

The speed control for the PMSM-BLAC motor was achieved with an external loop, over the previously mentioned current loop. The implemented controller was a PI as in the previous section to avoid issues arising due to the derivative component. In this sense, the noise introduced by the power electronic devices during the switching phase may have negative effects on the derivative component in any controller architecture.

PI control was carried out by simplifying the mechanical motor model, without considering the viscosity coefficient [9,28–30], yielding the open loop transfer function, where the PI constant for the speed control loop has an s subindex, as shown in Figure 4.

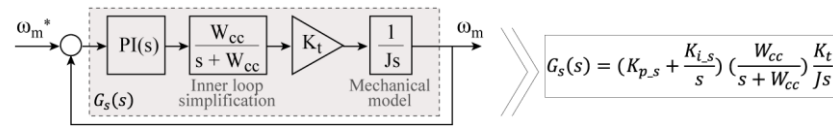


Figure 4. Simplified model of the external speed loop.

It is relevant to mention that in cascaded systems, the inner control variable has a larger bandwidth than the outer one, which results in a difference in response speed, where the inner variable is the faster one [26,30]. In this sense, the tuning bandwidth for the speed control is smaller than the one of the current control. Under the assumption that the cutoff frequency of the speed controller given by $W_{PI} = \frac{K_{i,s}}{K_{p,s}}$ is significantly smaller than the current control loop bandwidth W_{CC} , then the open loop transfer function can be approximated using the Bode diagram, as shown in Figure 5.

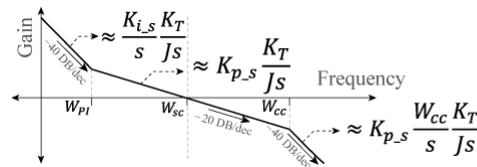


Figure 5. Bode diagram of speed control transfer function in open loop.

$G_s(s)$ can be approximated based on the dominant term, i.e., if $W < W_{PI}$, the integral term becomes dominant, yielding $G_s(s) \approx \frac{K_{i,s}}{s} \frac{K_T}{J s}$. The approximation for each region is listed in Figure 5. Since W_{PI} should be lower than W_{sc} , we can select a relationship of $W_{PI} = W_{sc}/4$, and with the aid of the Bode diagram in Figure 5, the PI constants can be computed as follows.

$$K_{p,s} = \frac{J W_{sc}}{K_T} \quad \& \quad K_{i,s} = K_{p,s} \frac{W_{sc}}{4}. \quad (11)$$

It is worth mentioning that the speed control will include an anti-wind-up configuration to avoid saturations in the integral component.

As mentioned before, field weakening allows for higher velocities than the nominal one to be reached. In order to prevent damage to the motor during the field weakening, we must apply two constraints: $i_d^2 + i_q^2 \leq I_{max}^2$ and $V_d^2 + V_q^2 \leq V_{max}^2$. This represents a circle in the $dq0$ frame [9,31,32]. Combining the decoupling expressions in Figure 3 with these constraints, it is possible to determine the current limits to prevent damage when applying field weakening, which leads to

$$i_d \leq \frac{\left(\frac{V_{max}}{w_e}\right)^2 - \Psi_m^2 - (L I_{max})^2}{2L \Psi_m} \quad (12)$$

2.4. Vector Space Pulse Width Modulation

Vector space PWM (SVPWM) allows for the minimization of harmonic distortion, delivering a higher voltage compared to other techniques and also reducing the losses produced by switching. In this modulation technique, any voltage vector V_{ref} can be computed from V_α and V_β , which are voltage components in the $dq0$ frame estimated using Park inverse transform. The angle α allows us to identify the sector number of any voltage reference, where it can be decomposed using adjacent commutation configurations, i.e., V_1 and V_2 in sector 1 and the null configuration V_0 , as shown in Figure 6 [8,9,33,34]. Figure 6 indicates how to compute the duration of each configuration, where the total commutation

time is $T_s = T_1 + T_2 + T_0$, n stands for the sector number that yields Equation (13), and the commutation configurations for other sectors can be found in [33].

$$\int_0^{T_s} \bar{V}_{ref} dt = \int_0^{T_1} \bar{V}_1 dt + \int_{T_1}^{T_1+T_2} \bar{V}_2 dt + \int_{T_1+T_2}^{T_s} \bar{V}_0 dt \quad (13)$$

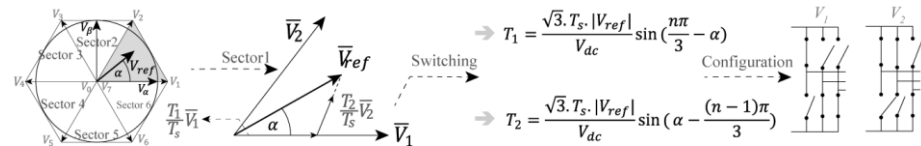


Figure 6. Voltage vector through different SVPWM sectors. Sector 1 example.

3. Results and Discussion

The proposed simulation model was implemented in MATLAB/SIMULINK using the motor parameters from Table 1, and it was evaluated in different scenarios, like the response of the current controller with different load profiles, including speeds over the nominal one, and locked rotor tests to assess dynamic changes in the setpoint of the current. It is worth mentioning that the switching frequency used in all tests was 10 kHz, and the current controller was assessed with different bandwidths, whose values were related to their constants according to Equations (10) and (11).

3.1. Internal Current Control Loop Tests

A locked rotor test ($J \rightarrow \infty$) allows us to assess the current loop, where we test different current controller bandwidths of f_{cc} ($\omega_{cc} = 2\pi f_{cc}$) of 500, 800, and 1000 Hz, and estimate the currents I_d and I_q settling time and error, as shown in Figure 7b. The computed PI constants allow for a reduced settling time of about 25 ms with mean current of zero in steady state. The controller performs better at higher current controller bandwidths, as shown in Table 2.

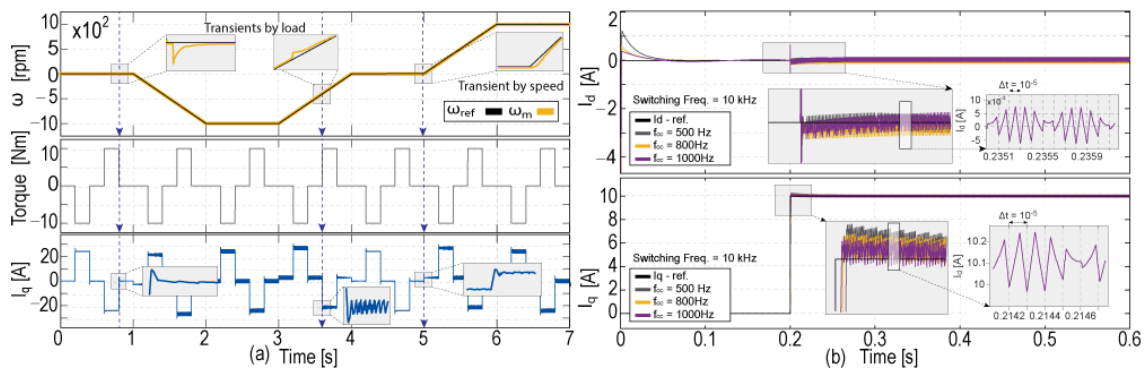


Figure 7. Control loop tests: (a) speed control, (b) current control in locked rotor.

Table 2. Performance benchmarking results for locked rotor and speed control experiments.

Locked Rotor Test					Speed Control Test						
Fcc (Hz)	Id		Iq		Fcc (Hz)	Speed		Id		Iq	
	ISE	Settling Time	ISE	ISE		ISE	Settling Time	ISE	Settling Time	ISE	Settling Time
500	2.68	35 ms	0.018	15 ms	800	0.14	120 ms	2.52	25 ms	0.012	11 ms
800	2.52	25 ms	0.012	11 ms	1000	0.125	103 ms	2.5	20 ms	0.009	8 ms
1000	2.5	20 ms	0.009	8 ms							

3.2. Velocity Control Loop Tests

The speed control loop makes use of a speed profile that remains unaffected by torque profile disturbance, as shown in Figure 7a. For visualizing purposes, we highlighted the transients during the load or velocity changes (see Figure 7a). This test was carried out with two current controller bandwidths of 800 and 1000 Hz. The error results are detailed in Table 2.

3.3. System Tests including Field Weakening

In this experiment we used the best-achieving current controller bandwidth of 1000 Hz and set a reference speed over the nominal one (1590 rpm). With no field weakening, the speed becomes saturated at a value closer to the nominal one (see yellow line in Figure 8a). Applying the field weakening constraint in Equation (13), the motor can exceed the nominal speed without surpassing the maximum current, retaining the nominal power, as shown in the orange plot in Figure 8a and the current plot on Figure 8b.

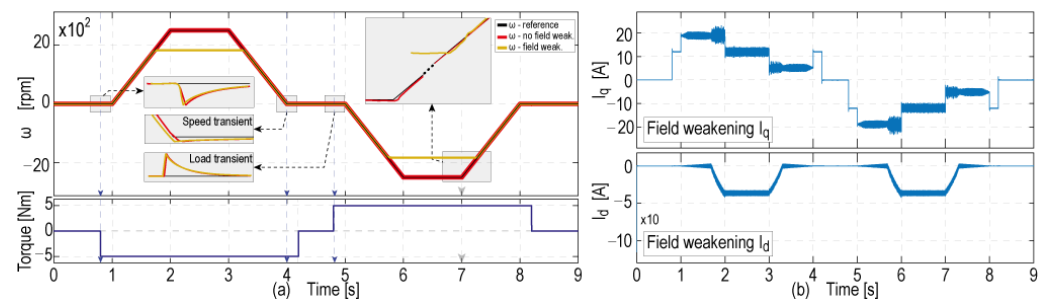


Figure 8. System response with field weakening: (a) speed response; (b) current response.

4. Conclusions

This study emphasizes the importance of the implementation of FOC in conjunction with field weakening in permanent-magnet synchronous machines. This approach grants these motors the capability to operate effectively across a wide range of speeds, offering a commendable torque response at low speeds while maintaining robust performance at high velocities.

Through the correct implementation of FOC with current and speed control, it can be ensured that a salient-pole PMSM has a stable response to speed and load profiles, with low response times and reduced over-peaks, and by using SVPWM in the three-phase inverter bridge, the available DC bus can be used in a more efficient way.

During the speed test, it was found that the integrity of control remained intact amidst dynamic shifts in speed setpoints and the introduction of load perturbations. This robustness was maintained as long as the specified speed references adhered to the desired values, irrespective of whether they fell below or soared above the nominal threshold. Notably, the utilization of field weakening played a critical role in ensuring control resilience across velocities surpassing the nominal threshold.

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